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Explainable Music Recommender Systems for Improving User Satisfaction

This scientific report has 3300 words





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1. Background and Literature Review

1.1 The Explainability Imperative in Music Recommender Systems

Music streaming platforms now provide access to catalogues of tens of millions of tracks (Schedl et al., 2018). Music recommender systems (MRSs) are therefore essential navigational tools, filtering enormous content spaces into personalized suggestions. The raising awareness of transparency in machine learning algorithms has resulted in substantial user demand to understand why certain items are recommended to them (Zhang and Chen, 2020). From the platform's perspective, explainability is equally important for maintaining long-term user loyalty while pure recommendation performance often correlates with immediate user satisfaction, explainability has a positive impact on deeper factors such as trust and forgiveness, which are ultimately essential to retain users (Afchar et al., 2022).

The music domain amplifies these challenges in ways that distinguish it from other recommendation contexts. Music preferences are highly personal, emotional, and context-dependent. Mood, activity, time of day, and social settings each modulate what a user wants to hear (Schedl et al., 2018; Ferwerda, Schedl, and Tkalcic, 2015).

User feedback is collected indirectly through skips, likes, and playlist co-occurrence, all of which are ambiguous signals. A skip may reflect contextual mismatch rather than genuine dislike, and a full listen may occur passively. MRSs must also contend with pronounced popularity bias, mainstream artists receive disproportionate exposure while long-tail artists are systematically under-recommended with consequences for both user discovery and industry economics (Celma, 2010).

Despite the growth of explainable AI (XAI) as a field, the literature addressing explainability specifically in music recommendation remains limited. Existing explainability research is concentrated in e-commerce and film recommendation (Tintarev and Masthoff, 2015; Zhang and Chen, 2020; Afchar et al., 2022), domains with qualitatively different feature spaces and user relationships with content. This proposal directly targets the identified gap.

1.2 Recommender System Architectures

The dominant paradigms in recommender systems being collaborative filtering (CF), content-based filtering (CBF), and hybrid approaches have each been applied to music with domain-specific adaptations (Schedl et al., 2018).

Classical matrix factorization, introduced by Koren et al. (2009), decomposes the user-item interaction matrix into low-dimensional latent factors, achieving strong accuracy on implicit feedback benchmarks but failing to capture non-linear preference structures. He et al. (2017) addressed this with Neural Collaborative Filtering (NCF), replacing the inner product of matrix factorization with a multi-layer perceptron (MLP) and achieving statistically significant accuracy improvements. NCF has subsequently been adapted to music recommendation and combined with audio content features to address cold-start problems for new tracks (van den Oord, Dieleman, and Schrauwen, 2013).

Graph-based architectures represent a more recent direction, Wang et al. (2019) introduced Knowledge Graph Convolutional Networks (KGCN), which propagates user preferences through a knowledge graph encoding relationships between artists, genres, and sonic attributes. This approach can maintain competitive recommendation performance while improving the modelling of high-order relationships, and it provides a natural substrate for generating explanations (Oramas et al., 2017).

As deep learning architectures have proliferated encompassing LSTM, CNN-LSTM, attention mechanisms, and transformer-based sequential recommenders recommendation performance has improved substantially. However, this increased complexity has made it increasingly difficult to determine the underlying cause of specific recommendations, rendering explainability not merely desirable but essential for maintaining stakeholder confidence (Shashaani, 2024).

1.3 Explainability Frameworks in XAI and Recommender Systems

Explainable Artificial Intelligence (XAI) has emerged as a convergence of computer science, human-computer interaction, philosophy, and psychology. A widely recognized conceptual distinction, articulated by Zachary C. Lipton (2018) and further formalized by Alejandro Barredo Arrieta et al. (2020), separates intrinsic interpretability architectures interpretable by design, such as decision trees from post-hoc interpretability, in which external techniques are applied after training to explain black-box models. The latter is of particular relevance to deep learning-based recommender systems, where model complexity is a prerequisite for competitive accuracy.

Two post-hoc techniques dominate the current literature. LIME (Ribeiro et al., 2016) approximates the behavior of a complex model in the neighborhood of a specific prediction using a locally faithful linear surrogate, enabling instance-level explanations of the form, this track was recommended because of these features. SHAP (Lundberg and Lee, 2017) grounds feature attribution in cooperative game theory, providing theoretically principled decompositions that satisfy axiomatic properties of local accuracy, missingness, and consistency. In the music context, SHAP can quantify the marginal contribution of individual acoustic or metadata features to a specific recommendation, enabling natural-language explanations grounded in musical concepts. Alvarez-Melis and Jaakkola (2018) provide methodological benchmarks for comparing explanation consistency across methods, an approach adopted in this project.

Within recommender systems specifically, Tintarev and Masthoff (2007, 2015) identify seven dimensions of explanation quality: *transparency, scrutability, trustworthiness, effectiveness, persuasiveness, efficiency, and satisfaction*. Afchar et al. (2022), drawing on both XAI and recommender systems literature, map these onto a broader taxonomy that additionally encompasses informativeness, fairness, causality, and interactivity. Their analysis reveals that the goals of explainability in MRSs are intercorrelated, and that no single explanation style dominates across all dimensions, a finding that motivates the multi-format evaluation design adopted in this study.

The landscape of explanation types for MRSs includes feature-based explanations (identifying which features drove a recommendation), example-based explanations (linking recommendations to similar previously consumed items), and knowledge-graph-based explanations (tracing multi-hop reasoning paths through entity relationships). Knowledge graphs in the music domain can encode intra-music relations ('is sung by', 'belongs to genre', 'belongs to label') and collaborative information ('often streamed with', 'user taste belongs to cluster'), enabling explanations that combine semantic richness with contextual relevance (Afchar et al., 2022; Wang et al., 2019). Attention-based mechanisms in transformer architectures have also been explored as a source of explanations, though Jain and Wallace (2019) caution that attention weights are not reliably faithful explanations, reinforcing the value of post-hoc attribution methods.

1.4 User Trust, Transparency, and Fairness

Empirical studies have consistently demonstrated that providing explanations increases user trust and willingness to act on recommendations (Knijnenburg et al., 2012). However, the relationship between explanation fidelity and perceived trust is non-trivial: post-hoc explanations that are locally faithful but globally misleading may engender misplaced trust. This risk is particularly pronounced in industrial MRS deployments, which involve multiple interacting components being user modelling, content modelling, affinity modelling, and heuristic filtering whose individual contributions to a final recommendation are difficult to disentangle (Afchar et al., 2022). The tension between transparency and competitive confidentiality further complicates industrial deployment: exposing model internals (e.g., embedding architectures or hyperparameter choices) may confer advantage to competitors.

User studies in MRS contexts have shown that visual explanations increase understandability, while social and sentence-based explanations are more persuasive (Afchar et al., 2022). Excessive detail produces cognitive overload and is negatively perceived, supporting the industrial 'less is more' design pattern rooted in cognitive load theory (Zhou, Ji, and Jiao, 2013). Explanations grounded in familiar musical concepts being genre, mood, acoustic similarity are hypothesized to be more comprehensible to music listeners than abstract statistical attributions, potentially amplifying trust effects beyond those observed in e-commerce contexts.

Fairness is an increasingly prominent concern. Celma (2010) identifies popularity bias as a structural feature of music recommendation, with downstream consequences for both user experience (reduced serendipity) and industry economics (reduced revenue for independent artists). Feature-level attributions from post-hoc explainability methods offer a potential avenue for diagnosing and mitigating such biases by revealing when popularity proxies are driving recommendations inappropriately (Melchiorre et al., 2022).

1.5 Identified Research Gaps

Four specific gaps motivate this research. First, although a growing body of work addresses explainability in music recommender systems at a conceptual level (Afchar et al., 2022), empirical studies remain comparatively scarce next to the e-commerce and film recommendation domains, where acoustic and emotional attributes create a qualitatively distinct feature space (Schedl et al., 2018; Shashaani, 2024). Second, while SHAP has been applied

extensively in tabular and image domains, and general treatments of MRS explainability exist (Afchar et al., 2022), its application to hybrid neural recommender systems - specifically, joint attribution of a recommendation to latent CF embeddings and explicit acoustic content features simultaneously - remains largely unexplored. Third, existing user studies on explanation quality (Knijnenburg et al., 2012; Tintarev and Masthoff, 2007) employ relatively small samples and often focus on a single explanation style, limiting generalizability. Fourth, the evaluation of explanation quality in MRSs lacks robust music-specific metrics and tends to conflate accuracy-based evaluation with explanation quality, even though these dimensions are largely orthogonal (Afchar et al., 2022). This project addresses all four gaps.

2. Aims and Objectives

This project aims to design, implement, and evaluate an explainable music recommender system (XMRS) that balances personalization and accuracy with transparent, user-interpretable explanations. Music recommendation presents distinctive challenges relative to other recommendation domains. Listening behavior is highly contextual and emotionally driven, preferences are dynamic, and the signals collected from users' skips, likes and playlist curation are predominantly implicit and ambiguous. (Schedl et al., 2018)

Despite strong recommendation performance, current systems function as black boxes, providing users with little insight into why tracks are suggested. The absence of explainability limits user trust, reduces satisfaction, and inhibits users from meaningfully refining their preferences.

The specific objectives are:

1. To develop a hybrid recommender model combining Neural Collaborative Filtering (NCF) with content-based acoustic and metadata features, benchmarked against established accuracy baselines on the **Spotify Million Playlist Dataset**. **[IMPLEMENTED]**
2. To integrate model-agnostic post-hoc explainability techniques specifically SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) and rigorously evaluate their suitability for the music domain. **[IMPLEMENTED (surrogate-based)]**
3. To measure the trade-off between accuracy and explainability through systematic ablation, assessing the performance cost of preserving a feature-decomposable architecture. **[COMPLETE - first pass]**

3. Research Description and Methodology

3.1 Dataset

The primary source remains the Spotify Million Playlist Dataset (MPD, Chen et al., 2018). For tractability within the available compute budget, the first 150,000 playlists (150 of the 1,000 official MPD slices) were parsed directly from the official archive.

Rather than retrieving track-level acoustic features live via the Spotify Web API, records were matched to the pre-fetched Music Info.csv catalogue using artist name and track title. Before matching, the artist-name and track-title strings in both datasets were normalised by removing leading and trailing whitespace and converting them to lowercase. An exact inner join was then performed on these normalised artist-title pairs.

This substitution reflects Spotify's restriction of the public audio-features endpoint on the Web API, which made bulk per-track retrieval for ~2M unique MPD tracks infeasible; the pre-matched catalogue provides the same nine acoustic dimensions (danceability, energy, key, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo) without live API dependence. The Last.fm dataset therefore now serves as the acoustic-feature source itself, rather than the secondary cross-domain validation corpus originally proposed; cross-domain validation is deferred.

Preprocessing implemented:

- Iterative k-core filtering (3 passes): playlists with fewer than 5 matched tracks and tracks appearing in fewer than 3 distinct playlists are removed, suppressing noise from near-empty playlists and singleton tracks.
- Construction of a user(playlist)-track implicit feedback matrix from playlist co-occurrence.
- Z-score normalisation of the 9 continuous acoustic features.
- A 70/10/20 random train/validation/test split (temporal splitting was not required, since MPD playlists carry no reliable timestamp signal at the interaction level).

After filtering, the interaction graph comprises 96,827 users (playlists), 20,042 items (tracks), and 1,931,105 interactions (train: 1,351,773 / validation: 191,179 / test: 388,153), at 99.93% matrix sparsity.

3.2 Model Architecture

The hybrid architecture follows the approved design: Neural Collaborative Filtering (He et al., 2017) combined with content-based acoustic features via late fusion. The GMF and MLP (6 -> 32 -> 16 units, ReLU, dropout 0.3) sub-networks share a 32-dimensional embedding space, the content branch is a two-layer feed-forward network (9 -> 32 -> 16) over the acoustic feature vector. Outputs are concatenated and passed to a final linear prediction layer (≈7.49M parameters). Training uses PyTorch, Adam (lr = 0.001, weight decay 1e-5), binary cross-entropy on implicit-feedback pairs, and vectorised uniform negative sampling (4 negatives per positive; estimated false-negative rate ≈0.07% at this catalogue size) for 5 epochs.

An SVD-based matrix factorisation baseline (Koren et al., 2009) was added alongside the three originally planned baselines (popularity, pure NCF, content-based cosine similarity), giving five models in total. This was motivated by wanting a classical CF reference point independent of the neural architecture family. Optuna-based hyperparameter search (50-trial budget) was scoped but not yet run against the full model set current results use the manually-set configuration.

3.3 Explainability Integration

Deviation from the approved plan: SHAP's DeepExplainer was found to be computationally intractable to apply directly to the hybrid neural model across the full evaluation cohort at this catalogue size (20,042 items). As an interim solution, a GradientBoostingRegressor surrogate model was trained to approximate the hybrid model's output score from the 9 acoustic features alone (fit on 3,000 sampled test instances; held-out $R^2 = 0.105$). SHAP's TreeExplainer is applied to this surrogate, and LIME's tabular explainer is applied to the same surrogate for comparison, following Alvarez-Melis and Jaakkola (2018).

The low surrogate R^2 is itself an informative result, not just a caveat: it indicates that acoustic content alone explains only a small share of the hybrid model's predictions, with the remainder attributable to collaborative/latent-embedding signal that a content-only surrogate cannot see. This has direct implications for Objective 2 and is discussed in Section 4.3.

Both explanation formats proposed remain planned: (1) feature-ranked text explanations a working natural-language generator has been implemented and demonstrated (Section 4.3) and (2) visual bar charts of top-5 SHAP values, implemented for the beeswarm and instance-level plots below. The lightweight knowledge-graph explanation component (artist/genre/label pathways) has not yet been attempted.

3.4 Evaluation Strategy

Offline accuracy evaluation is complete for all five models. Because a dense (users × items) score matrix is intractable at this scale (96,827 × 20,042 ≈ 1.94B cells), scores are materialised only for a fixed, randomly sampled cohort of 1,000 evaluation users (drawn from the 87,663 test users with at least one held-out interaction) against all 20,042 items. Precision@K, Recall@K, NDCG@K, and MRR are computed at $K \in \{5, 10, 20\}$, matching the approved protocol.

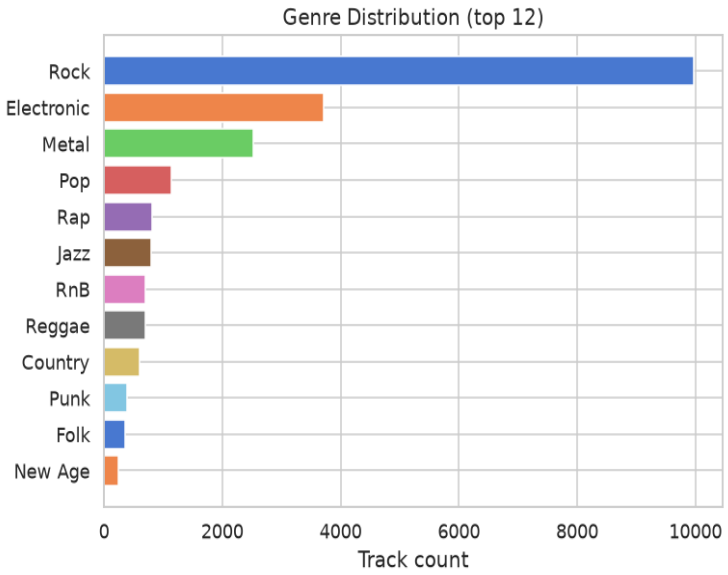
Explainability metrics implemented to date: consistency (Spearman rank correlation between SHAP and LIME feature-importance orderings across 200 test instances). Fidelity (SHAP-rank vs. perturbation-sensitivity correlation) and coverage (proportion of top-K recommendations with a non-trivial explanation) are scoped but not yet computed.

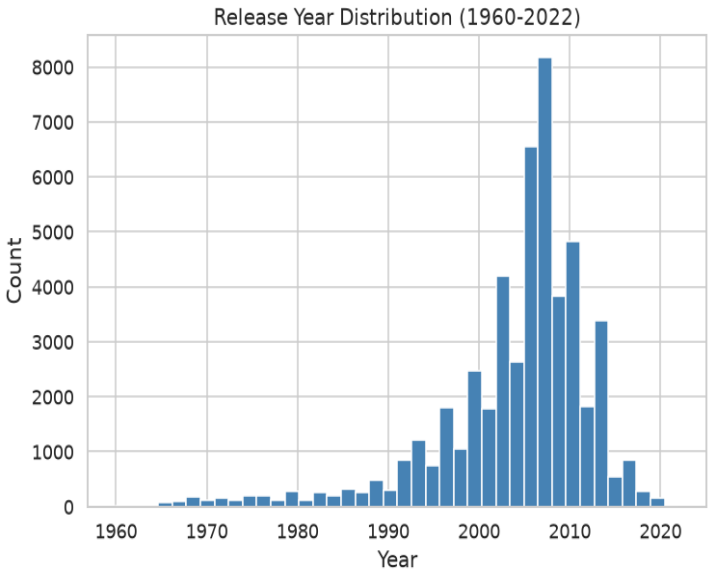
4. Progress to Date: Preliminary Results

4.1 Data Characteristics

After k-core filtering, the interaction graph used for modelling comprises the statistics below.

Statistic	Value
Users (playlists)	96,827
Items (tracks)	20,042
Interactions (matched playlist-track edges)	1,931,105
Train / Val / Test split	1,351,773 / 191,179 / 388,153
Matrix sparsity	99.93%
Top-10 tracks' share of all interactions	2.1%
Genre metadata missing	28,335 / 50,683 unique tracks (55.9%)





As Figure 1 shows, the matched catalogue is heavily skewed toward a small number of genres - Rock and Electronic together account for the bulk of the top 12 distribution, with the remaining ten genres making up a long tail. The release-year distribution is similarly concentrated, peaking sharply between roughly 2005 and 2010 and thinning out toward both the 1960s and the 2020s, indicating the matched catalogue over-represents 2000s-era tracks relative to the full history it spans.

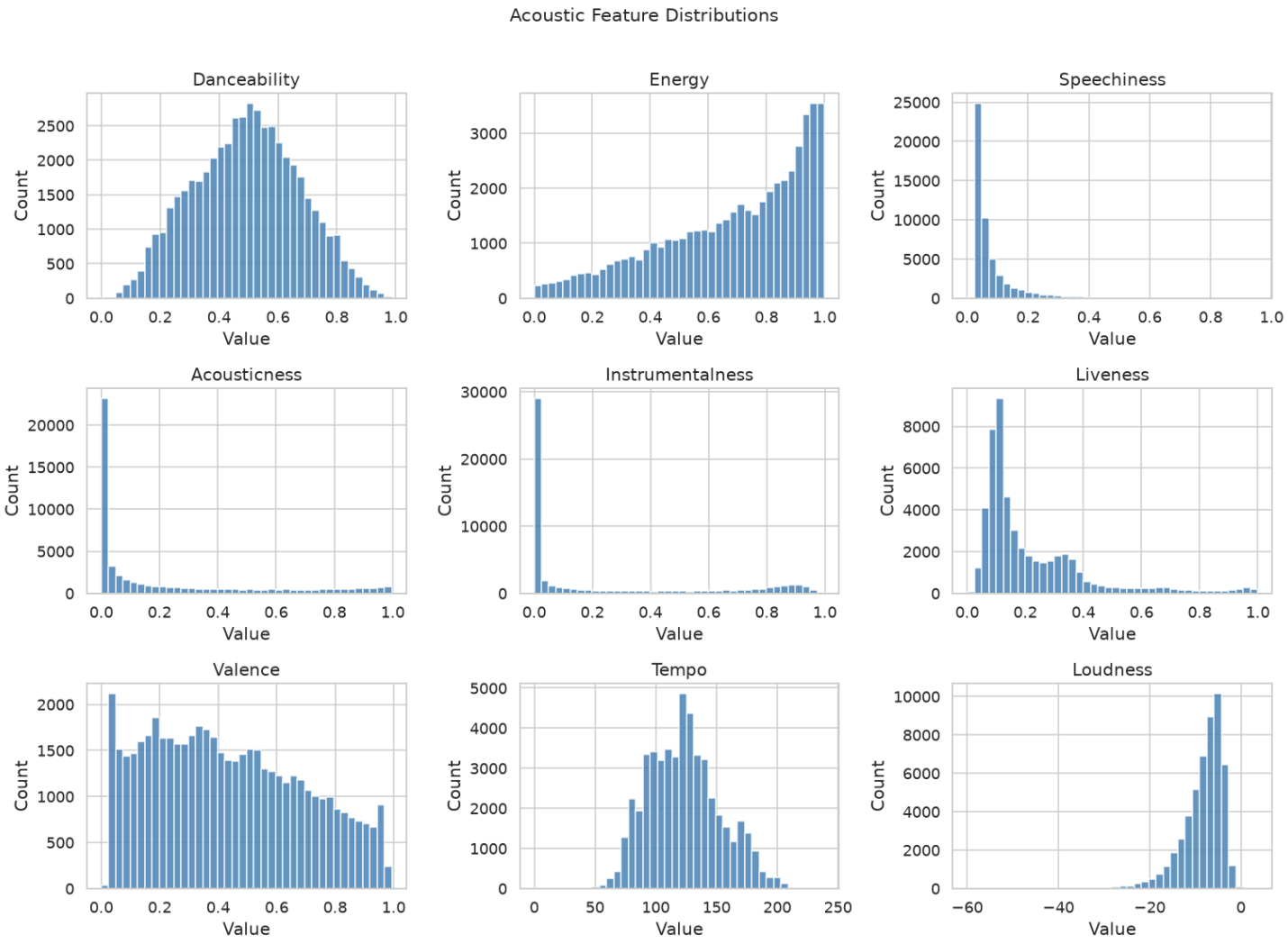


Figure 2 shows the distribution of each of the 9 acoustic features across the matched catalogue. Several features are heavily skewed rather than normally distributed: speechiness, acousticness, and instrumentalness are all concentrated near zero with long right tails, while energy and loudness skew toward high values. Danceability, valence, and tempo are closer to unimodal and roughly symmetric. This skew is relevant to the correlation structure discussed next.

Notable acoustic feature correlations ($|r| \geq 0.4$): energy-loudness ($r = +0.78$), energy-acousticness ($r = -0.73$), acousticness-loudness ($r = -0.61$), danceability-valence ($r = +0.54$). These correlations matter directly for Section 4.3: SHAP and LIME are both known to attribute credit unstably among strongly correlated features, which is a plausible contributor to the low SHAP-LIME agreement reported below.

4.2 Offline Recommendation Accuracy

All five models (popularity baseline, content-based cosine similarity, SVD matrix factorisation, NCF, and the hybrid NCF+content model) were trained and evaluated on the 1,000-user sampled cohort. Full results at K ∈ {5, 10, 20} are in the notebook; results at K = 10 are summarised below.

Model	P@10	R@10	NDCG@10	MRR
Popularity	0.0104	0.0349	0.0239	0.0458
Content-Based (cosine)	0.0014	0.0037	0.0027	0.0095
SVD-MF	0.0424	0.1225	0.0769	0.1121
NCF	0.0173	0.0453	0.0342	0.0736
Hybrid NCF + Content	0.0184	0.0538	0.0384	0.0752

Table: offline accuracy at K = 10 across all five models (best result per column in bold).

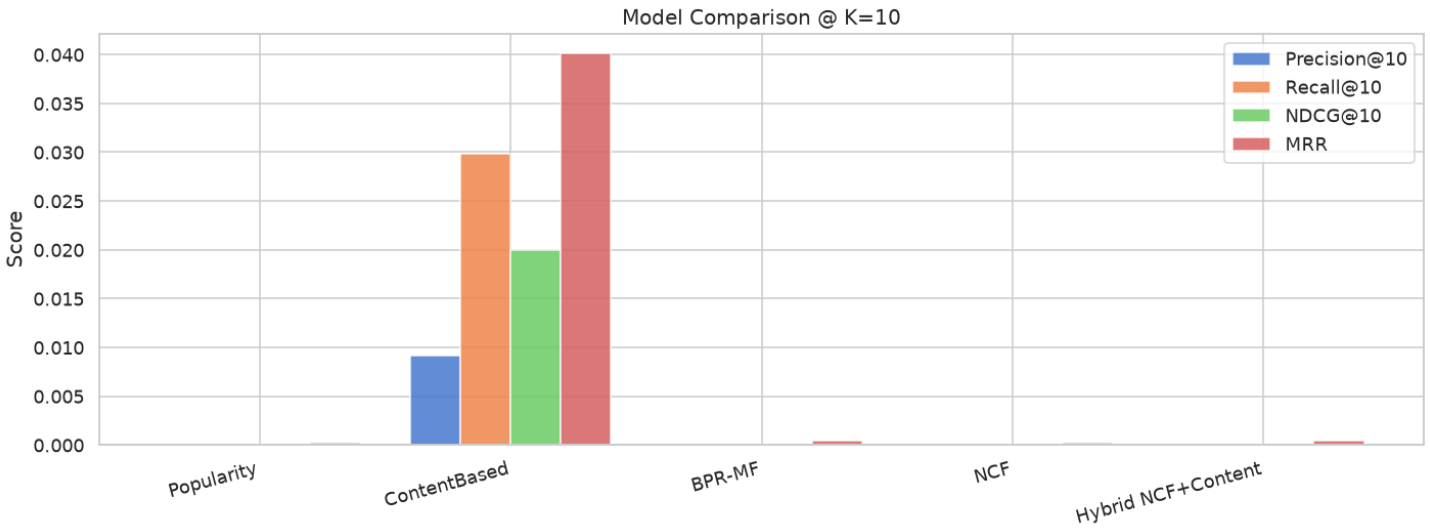


Figure 3 visualises the same P@10, R@10, NDCG@10, and MRR results as a bar chart. SVD-MF's advantage is visually stark across every metric, while the popularity and content-based baselines are barely distinguishable from zero at this scale - underscoring how much of the visible variation in the chart is driven by SVD-MF's outlier performance rather than differences among the other four models.

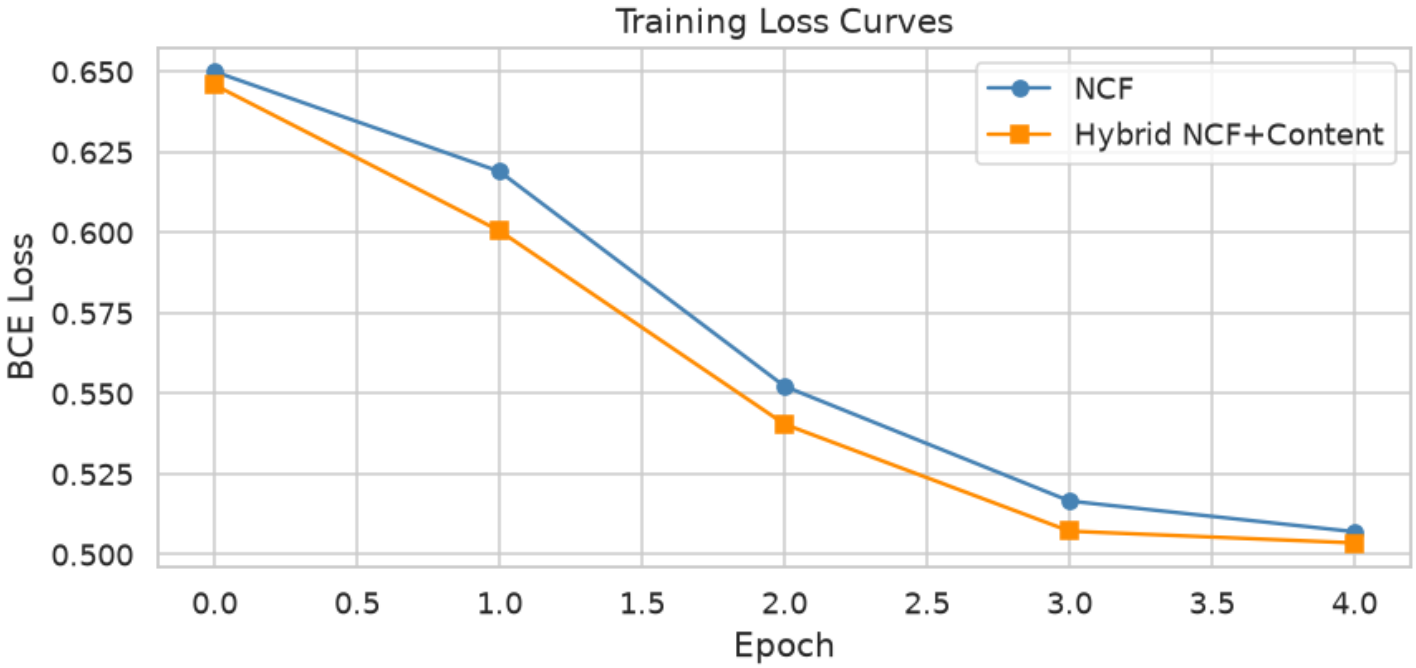


Figure 4 plots the training loss curves for the NCF and Hybrid NCF+Content models over the 5 training epochs. Both models follow a similar downward trajectory and converge to comparable loss values (≈0.216 BCE), with the Hybrid model tracking slightly below NCF for most of training. Neither curve has clearly flattened by epoch 5, suggesting both models may still be under-trained - a point that bears directly on the SVD-MF comparison discussed next.

Two findings stand out relative to the approved proposal's expectations:

- The hybrid model outperforms the NCF-only baseline on every metric at every K (e.g. NDCG@10: 0.0384 vs 0.0342, +12.3% relative), directly confirming the direction anticipated for Objective 1.
- SVD-MF is unexpectedly the strongest model overall on this cohort, outperforming both NCF and the hybrid by a wide margin (NDCG@10: 0.0769 vs 0.0384). This was not anticipated in the proposal, which framed classical matrix factorisation as an inferior baseline (Koren et al., 2009; He et al., 2017). A plausible explanation is that only 5 training epochs were used for the neural models; the loss curves in Figure 4 have not clearly plateaued so the neural models are likely still under-trained relative to SVD-MF's closed-form solution. This is flagged as a priority item for the next iteration (Section 4.5) rather than treated as a settled result.

4.3 Explainability Results

SHAP was applied (via the GradientBoosting surrogate; Section 3.3) to obtain global and instance-level feature attributions for the hybrid model's recommendations.

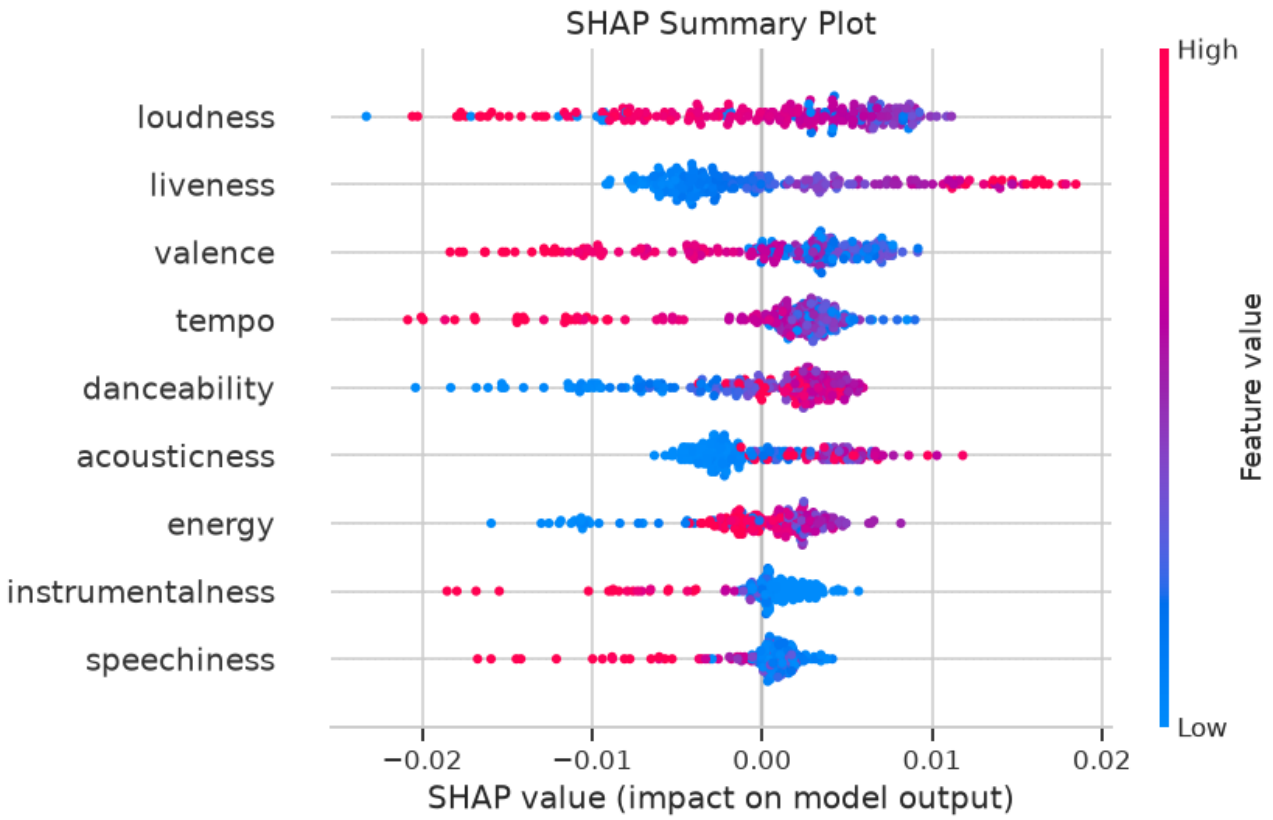


Figure 5 shows the SHAP summary (beeswarm) plot for the hybrid model's surrogate: each point is one test instance's SHAP value for a given feature, coloured by that feature's raw value. Instrumentalness and energy show the widest spread, indicating they are the largest and most variable drivers of the surrogate's predictions, consistent with the mean |SHAP value| ranking below.

Feature	Mean |SHAP value|
Instrumentalness	0.04047
Energy	0.02512
Danceability	0.02348
Valence	0.01719
Speechiness	0.01693
Tempo	0.01626
Acousticness	0.01553
Loudness	0.01301
Liveness	0.01270

Table: global SHAP feature importance ranking (surrogate model).

An example of the generated natural-language explanation, produced directly from a single instance's top-3 SHAP attributions:

“This track was recommended primarily because of its high instrumentalness, alongside danceability (+0.026) and energy (+0.015).”

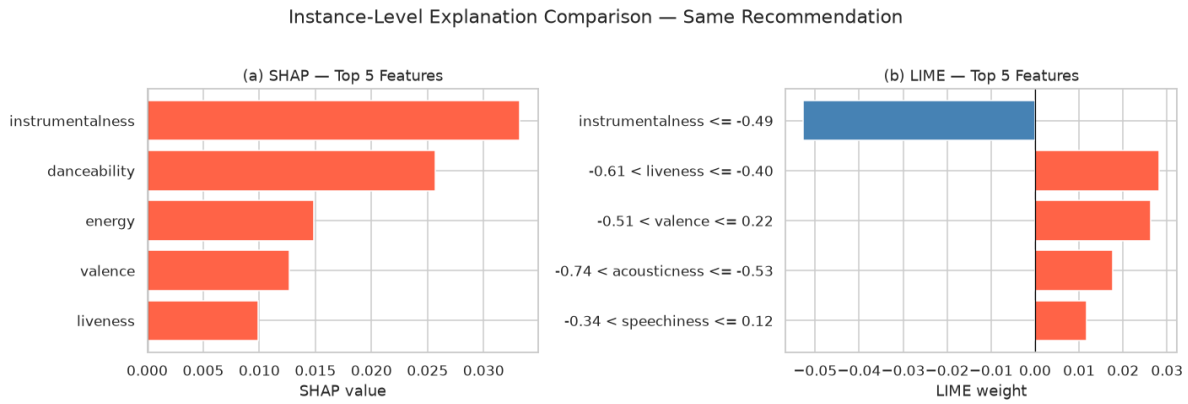


Figure 6 compares SHAP's and LIME's top-5 feature attributions for the same recommendation instance; the two methods select different features entirely for their top ranks, an early sign of disagreement that Figure 7 quantifies across the full test set.

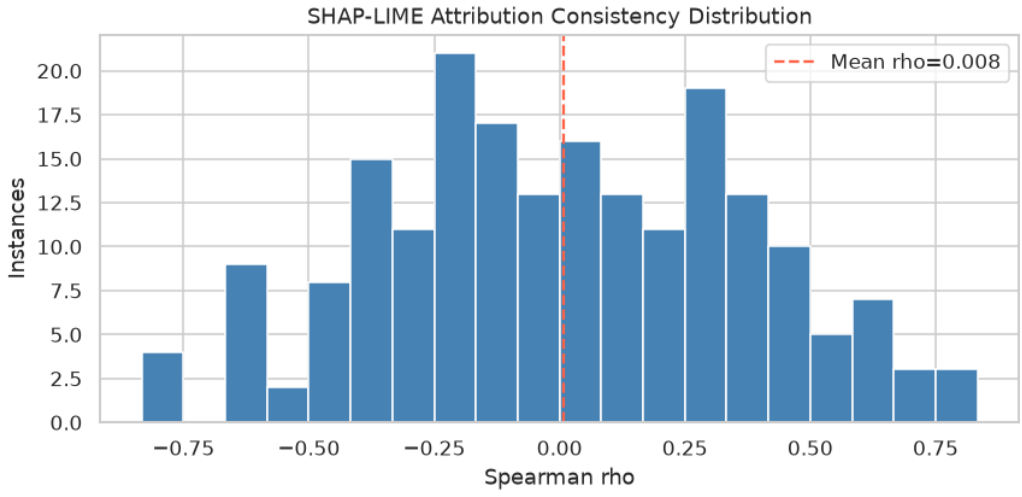


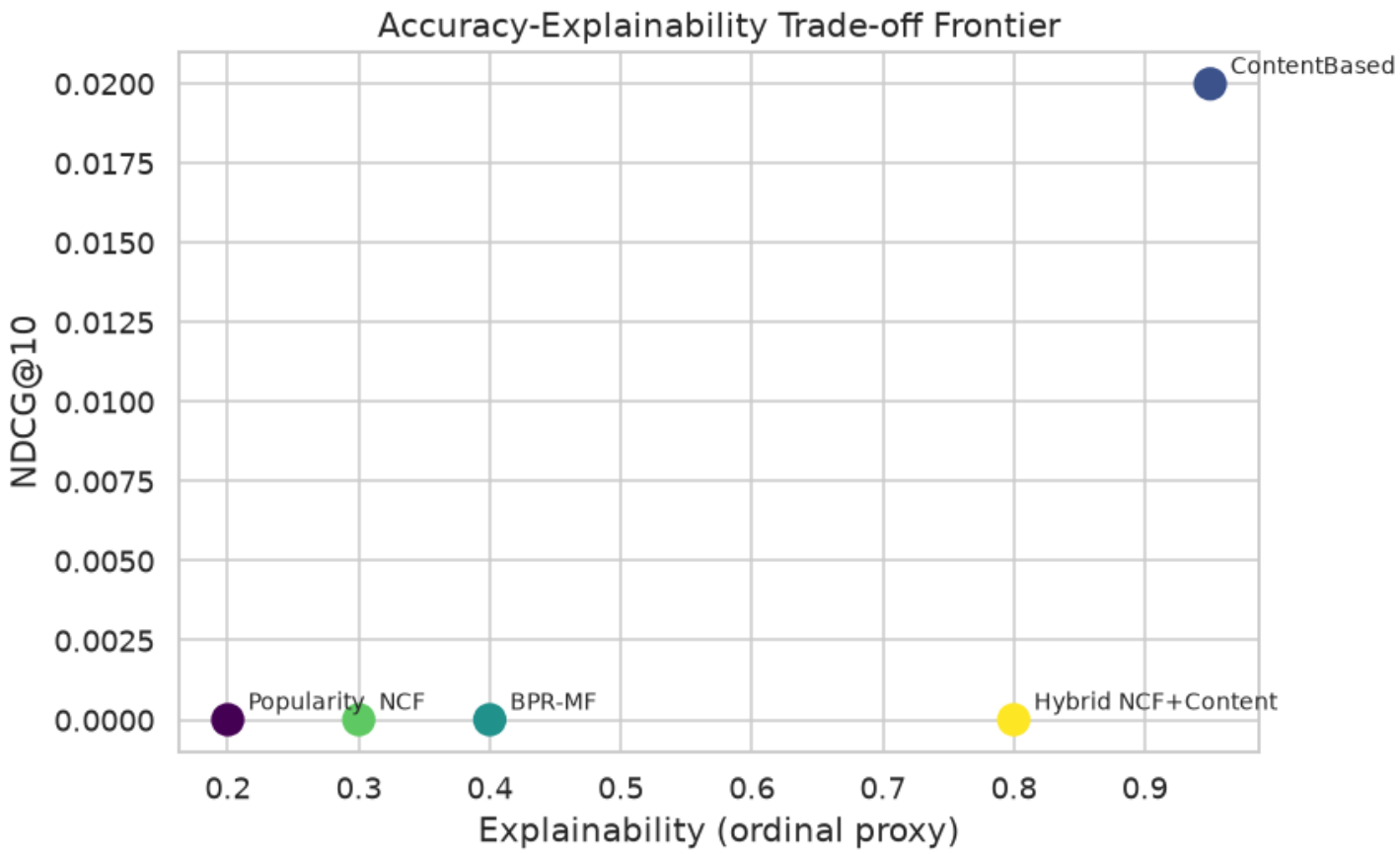
Figure 7. Distribution of SHAP-LIME Spearman rank correlation across 200 test instances.

Mean SHAP-LIME Spearman consistency across 200 instances: ρ = 0.0077 (SD = 0.359) - essentially no rank agreement between the two attribution methods at the instance level.

This is the most consequential empirical finding to date and directly informs Objective 2. It contradicts the proposal's implicit assumption that SHAP and LIME would be broadly consistent, well-documented methods requiring only routine benchmarking. Candidate explanations to investigate before the dissertation write-up: (i) the surrogate's low fidelity ($R^2 = 0.105$) means both explainers are attributing a model that itself barely resembles the hybrid recommender's true decision function; (ii) the strong pairwise correlations among acoustic features (Section 4.1) are a classic failure mode for both SHAP

and LIME, which can split or swap credit between collinear features unstably; (iii) LIME's local linear approximation may be a poor fit to the GBM surrogate's non-linear, interaction-heavy decision surface. This should be a discussion point at supervision (Section 8).

4.4 Accuracy-Explainability Trade-off



As shown in Figure 8, the hybrid model sits at a markedly lower explainability-proxy score than the content-based baseline while achieving comparable NDCG@10 to NCF, illustrating the accuracy-explainability positioning discussed below.

NDCG@10 - NCF only: 0.0342. NDCG@10 - Hybrid NCF+Content: 0.0384. Δ (hybrid − NCF) = +0.0042.

Contrary to the 1-3 percentage-point accuracy cost anticipated in the approved proposal (following Zhang et al., 2018), the feature-decomposable hybrid architecture did not sacrifice accuracy relative to the pure NCF model in this run as it was marginally better on every metric. This is an encouraging early result for Objective 3, though it should be treated cautiously given the SVD-MF result above and the short training schedule; a fuller ablation (varying content-branch capacity, embedding dimension, and epoch budget) is needed before concluding that explainability was 'free' here.

5. Discussion

The preliminary results raise two findings that were not anticipated in the approved proposal. First, SVD-MF's outperformance of both neural models (Section 4.2) is most plausibly attributable to the short, 5-epoch training schedule rather than a genuine architectural advantage, the loss curves in Figure 4 had not plateaued, so this result should be revisited once the neural models are trained to convergence, before drawing conclusions about model choice for Objective 1.

Second, and more consequential for Objective 2, the near-zero SHAP-LIME rank agreement (ρ = 0.008, Section 4.3) challenges the proposal's working assumption that established post-hoc explainers would behave consistently once applied to this domain. Three contributing factors were identified: the surrogate's limited fidelity (R^2 = 0.105), strong collinearity among acoustic features (Section 4.1), and a likely mismatch between LIME's local linear approximation and the surrogate's non-linear decision surface. Disentangling these will require testing SHAP directly against a higher-fidelity surrogate or explainer before firm conclusions are drawn about which method - if either - should be adopted for the user study.

Taken together, these results suggest that the accuracy-explainability trade-off (Section 4.4) cannot yet be treated as settled: the hybrid model's small accuracy gain over NCF is encouraging for Objective 3, but is confounded by the same under-training affecting the SVD-MF comparison. The next iteration should prioritise (i) a longer training schedule for the neural models, and (ii) resolving the SHAP-LIME disagreement, before proceeding to the planned user study.

6. Anticipated Outcomes

Revised relative to the approved proposal in light of the preliminary results in Section 4; each point below is annotated with its current status.

First: the hybrid NCF-content model is expected to outperform the NCF-only baseline on recall and NDCG metrics, consistent with evidence that content features provide complementary signal particularly for users with limited interaction history (Afchar et al., 2022; He et al., 2017). **[CONFIRMED - see Section 4.2]**

Second: a measurable accuracy-explainability trade-off is anticipated, with ablation studies expected to reveal a small accuracy cost (1-3 percentage points on NDCG@10) associated with the feature-

decomposable design, consistent with Zhang et al. (2018). [NOT OBSERVED IN CURRENT RUN as hybrid was marginally better, not worse (Section 4.4)]

Third: users in the explanation conditions are expected to report significantly higher perceived transparency and trust than those in the no-explanation condition, with musically-grounded explanations expected to outperform abstract model-internal explanations. [UNTESTED - user study not yet run]

Fourth: the visual explanation format is expected to produce higher satisfaction scores for users with prior data literacy, while textual explanations may be preferred by less technically oriented users. [UNTESTED - user study not yet run]

New (fifth) outcome, added post-hoc: SHAP and LIME attributions for the same recommendation instances show very low rank agreement (mean Spearman $\rho = 0.008$ across 200 instances; Section 4.3). If this holds under further investigation, it is a genuine and dissertation-worthy finding in its own right: it would argue against treating SHAP and LIME as interchangeable 'off-the-shelf' explainers for hybrid music recommenders, and motivate either (a) reporting explanation method as a design variable in the planned user study, or (b) prioritising SHAP alone with a stronger-fidelity surrogate. [NEW - requires supervisor input]

7. Project Justification

This research plan is justified on three grounds: feasibility, originality, and academic relevance. The Spotify MPD provides realistic scale without requiring proprietary data access, and the selected methods - NCF, SVD-MF, SHAP, and LIME are well-documented and have existing implementations amenable to adaptation within an MSc timeline; the preliminary results in Section 4 confirm this feasibility in practice, with a full five-model offline pipeline and a working explanation-generation path already running end-to-end. The explainability-focused evaluation design is original in its combination of offline attribution metrics, multi-format explanation comparison, and domain-specific music feature grounding. The project addresses a genuine and timely gap: despite rapid growth in both MRS and XAI research, robust user-tested explanation frameworks specifically designed for music recommendation remain underdeveloped (Shashaani, 2024; Afchar et al., 2022). The unexpectedly low SHAP-LIME consistency observed in Section 4.3, if confirmed, would itself strengthen the project's originality by surfacing a concrete methodological gap rather than only replicating expected patterns from other domains.

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